



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech.(MMD) (CBCS) Degree Examinations
 January 2023

22MMD101 - COMPUTATIONAL METHODS IN ENGINEERING

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Any One full questions from Each Unit.

		Unit - I	Marks	BT*	CO*	PO*
1.	a)	$A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \end{bmatrix}$, then find the matrix X, such that $2A + 3X = 5B.$	5	L*2	1	1
	b)	$A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ Find the inverse of [A]	5	L2	1	1
	c)	Solve by Gauss Elimination method $5x_1 - x_2 + 2x_3 = 12$ $3x_1 + 8x_2 - 2x_3 = -25$ $x_1 + x_2 + 4x_3 = 6$	10	L2	1	1
2.	a)	$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$; Prove that $A^3 - 23A - 40I = 0$	5	L2	1	1
	b)	$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & 6 \\ -1 & 0 & -3 \end{bmatrix}$ Find the inverse of [A]	5	L2	1	1
	c)	Solve by Gauss Seidal method $5x_1 - x_2 = 9$ $-x_1 + 5x_2 - x_3 = 4$ $-x_2 + 5x_3 = -6$	10	L2	1	1
		Unit - II				
3.	a)	Solve the ODE given $4xy \, dx + (x^2 + 1) \, dy = 0$	5	L2	2	1
	b)	Solve the given ODE $y' = 4y + 2x - 4x^2$.	5	L2	2	1
	c)	Find complete solution of the PDE given, $r - 2s + t = \sin(2x + 3y)$	10	L2	2	1
4.	a)	Solve the given ODE $y' = \sin^2(x - y + 1)$	6	L2	2	1
	b)	Solve by Lagrange method: $z(z^2 + xy)(px - qy) = x^4$	8	L2	2	1
	c)	Obtain complete solution for 1D Heat equation as a Fourier series.	6	L2	2	1
		Unit - III				
5.	a)	Find the Fourier series of $f(x) = \begin{cases} 0 & \text{when } -\pi \leq x \leq 0 \\ x^2 & \text{when } 0 \leq x \leq \pi \end{cases}$	10	L1	3	1
	b)	Find Laplace Transform of $f(t) = \left(\sqrt{t} - \frac{1}{\sqrt{t}} \right)^3$	10	L1	3	1
6.	a)	Find the Fourier series of $f(x) = 2x - x^2$ in an interval of 0 to 3	10	L1	3	1
	b)	Find Laplace Transform of $(t^2 - 3t + 2) \sin 3t$	10	L1	3	1

Unit – IV

7. a) Find the approximate value of the real root of the equation $2x - \log_{10} x - 7 = 0$ by using Newton – Raphson method.
- b) Given $y = \log x$ and

x	4.0	4.2	4.4	4.6	4.8	5.0	5.2
y	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

Evaluate $I = \int_4^{5.2} \log x \, dx$ by (a) Trapezoidal rule; (b) Simpson's $\frac{1}{3}$ rule (c) Simpson's $\frac{3}{8}$ rule and compare it with exact value.

10 L2 4 1

8. a) Find a positive root of the equation $x^4 - x = 10$ by using Newton – Raphson method.
- b) The speeds of an electric train at various times after leaving one station until it stops at the next station are given in the following table. Find the distance between the two stations.

Speed in mph	0	13	33	39.5	40	40	36	15	0
Time in minutes	0	0.5	1	1.5	2	2.5	3	3.25	3.5

10 L2 4 1

10 L2 4 1

10 L2 4 1

Unit – V

9. a) A die is thrown 132 times with following results:

Number turned up	1	2	3	4	5	6
Frequency	16	20	25	14	29	28

Is the die unbiased?

10 L2 5 1

- b) Explain the general model of experiments and factorial experimental design.

10 L1 5 1

10. a) Find the value of χ^2 for the following information:

Class	A	B	C	D	E
Observed frequency	8	29	44	15	4
Theoretical (or expected) frequency	7	24	38	24	7

10 L2 5 1

- b) Explain the alternate method for One way ANOVA.

10 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
